

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458258.881 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1272 +/- 0.0055
Transit depth (Rp/Rs)^2	1.62 +/- 0.14 [%]
Orbital Inclination (inc)	88.01 +/- 0.48
Airmass coefficient 1 (a1)	1.2525 +/- 0.0065
Airmass coefficient 2 (a2)	-0.1685 +/- 0.0054
Scatter in the residuals of the lightcurve fit is	0.52 %
Best Comparison Star	None
Optimal Aperture	4.8
Optimal Annulus	9.99
Transit Duration (day)	0.1177 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1272 $\pm$ 0.0055 vs 0.1326 (deviation 0.95 σ)
Duration fit vs published	0.1177 d vs 0.1238 d (deviation 1.43 σ)
Mid-time O-C	6.1 min
Depth SNR	22.5
Residual scatter	0.52 %

Light curve

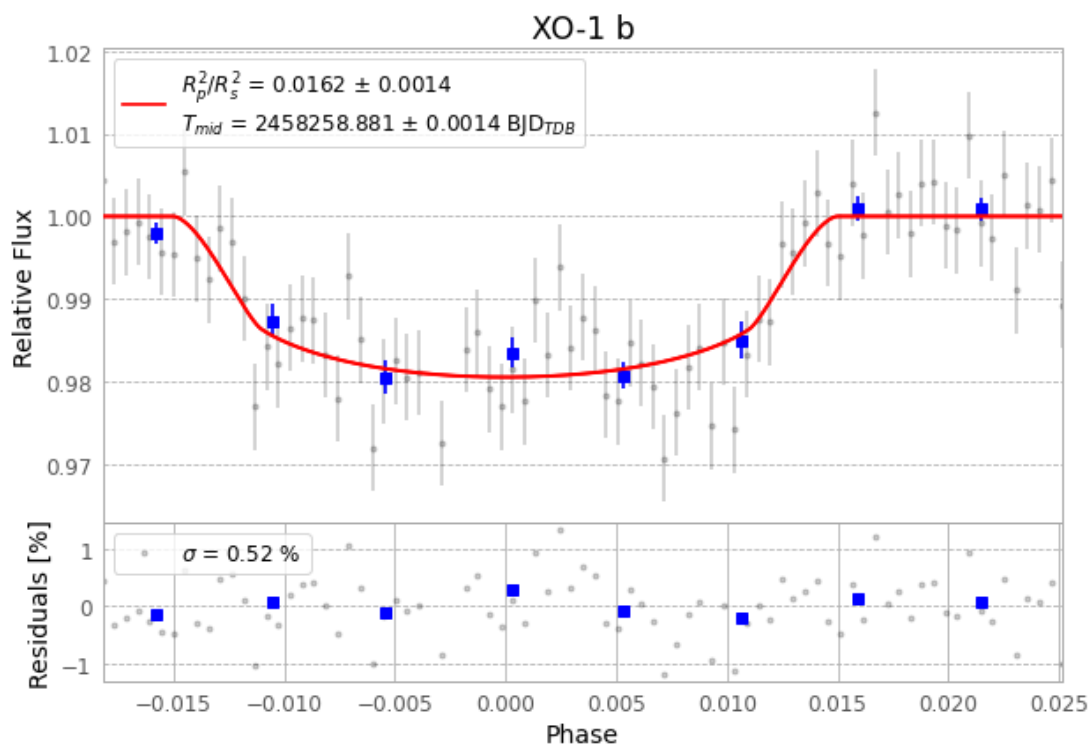


Figure 1 — FinalLightCurve_XO-1 b_2018-05-20.png (SHA-256 e01f4d1f5b3a2369...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 260], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b55347cccd9e4525...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

83 FITS frames (55 MB), XO-1180520071812.FITS → XO-1180520112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest.

Artifacts: bench-XO-1-180520.log (aa72760744c7...), FinalLightCurve_XO-1 b_2018-05-20.png (e01f4d1f5b3a...), FinalLightCurve_XO-1 b_2018-05-20.csv (c513e03a450a...)

Compute resources

Wall time 195.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 22:50Z.